

Solving Trigonometric Equations

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Note: All solutions in $[0, 2\pi)$ unless stated otherwise. General form given where applicable.



Mr. Kenny's Key Point

Two big traps in trig equations:

1. **Always check the range** ($|\sin| \leq 1$, $|\cos| \leq 1$) — extraneous roots like $\cos x = 3$ must be rejected.
2. For **multiple-angle** equations ($2x$, $x/2$, etc.), find ALL solutions in the wider domain first, then translate back to x .

Part A — Direct Equations & Factoring

1. $2 \sin x - 1 = 0$ on $[0, 2\pi)$

Isolate: $2 \sin x = 1 \Rightarrow \sin x = \frac{1}{2}$

Reference angle: $\frac{\pi}{6}$

Sine is positive in QI and QII: $x = \frac{\pi}{6}, \frac{5\pi}{6}$

$$x = \frac{\pi}{6}, \frac{5\pi}{6}$$

2. $\tan x + 1 = 0$ on $[0, 2\pi)$

Isolate: $\tan x = -1$

Reference angle: $\frac{\pi}{4}$

Tangent is negative in QII and QIV:

$$x = \pi - \frac{\pi}{4} = \frac{3\pi}{4}, \quad x = 2\pi - \frac{\pi}{4} = \frac{7\pi}{4}$$

$$x = \frac{3\pi}{4}, \frac{7\pi}{4}$$

3. $2 \sin \theta \cos \theta + \cos \theta = 0$ on $[0, 2\pi)$

Factor: $\cos \theta (2 \sin \theta + 1) = 0$

Case 1: $\cos \theta = 0 \Rightarrow \theta = \frac{\pi}{2}, \frac{3\pi}{2}$

Case 2: $2 \sin \theta + 1 = 0 \Rightarrow \sin \theta = -\frac{1}{2}$

Sine is negative in QIII and QIV:

$$\theta = \pi + \frac{\pi}{6} = \frac{7\pi}{6}, \quad \theta = 2\pi - \frac{\pi}{6} = \frac{11\pi}{6}$$

$$\theta = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}$$

4. $\sin^2 x - \cos^2 x = 0$ on $[0, 2\pi)$

Rewrite: $\sin^2 x = \cos^2 x$

Divide by $\cos^2 x$ (valid wherever $\cos x \neq 0$, checked separately):

$$\tan^2 x = 1 \Rightarrow \tan x = \pm 1$$

Reference angle: $\frac{\pi}{4}$. Tangent is ± 1 in all four quadrants.

$$x = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$$

Check $\cos x = 0$: $x = \frac{\pi}{2}, \frac{3\pi}{2}$ give $\sin^2 x - \cos^2 x = 1 \neq 0$. Not solutions.

$$x = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$$

Part B — Quadratic in a Trig Function

5. $2 \cos^2 x - 7 \cos x + 3 = 0$ on $[0, 2\pi)$

Let $u = \cos x$: $2u^2 - 7u + 3 = 0$

Factor: $(2u - 1)(u - 3) = 0$

$u = \frac{1}{2}$ OR $u = 3$ **REJECT: $|\cos x| \leq 1$**

Solve $\cos x = \frac{1}{2}$: Reference angle $\frac{\pi}{3}$. Cosine positive in QI and QIV.

$x = \frac{\pi}{3}, \quad x = 2\pi - \frac{\pi}{3} = \frac{5\pi}{3}$

$x = \frac{\pi}{3}, \frac{5\pi}{3}$

6. $2 \cos^2 \theta + \sin \theta = 1$ on $(0, 2\pi]$

Substitute $\cos^2 \theta = 1 - \sin^2 \theta$:

$2(1 - \sin^2 \theta) + \sin \theta = 1$

$2 - 2 \sin^2 \theta + \sin \theta = 1$

$-2 \sin^2 \theta + \sin \theta + 1 = 0 \Rightarrow 2 \sin^2 \theta - \sin \theta - 1 = 0$

Let $u = \sin \theta$: $(2u + 1)(u - 1) = 0$

Case 1: $\sin \theta = 1 \Rightarrow \theta = \frac{\pi}{2}$

Case 2: $\sin \theta = -\frac{1}{2}$. Reference angle $\frac{\pi}{6}$. Sine negative in QIII and QIV.

$\theta = \pi + \frac{\pi}{6} = \frac{7\pi}{6}, \quad \theta = 2\pi - \frac{\pi}{6} = \frac{11\pi}{6}$

Interval $(0, 2\pi]$: all three solutions are in range ($\theta = 2\pi$ — check:

$\sin 2\pi = 0 \neq -\frac{1}{2}$, so only 3 solutions).

$\theta = \frac{\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}$

7. $\tan^2 x = 3$ on $[0, 2\pi)$

Take square root: $\tan x = \pm\sqrt{3}$

Reference angle: $\frac{\pi}{3}$ (since $\tan \frac{\pi}{3} = \sqrt{3}$)

$\tan x = \sqrt{3}$: QI and QIII $\Rightarrow x = \frac{\pi}{3}, \frac{\pi}{3} + \pi = \frac{4\pi}{3}$

$\tan x = -\sqrt{3}$: QII and QIV $\Rightarrow x = \pi - \frac{\pi}{3} = \frac{2\pi}{3}, 2\pi - \frac{\pi}{3} = \frac{5\pi}{3}$

$$x = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}$$

Part C — Multiple-Angle Equations

8. $\cos 2x = -1$ on $[0, 2\pi)$

Let $u = 2x$. Since $x \in [0, 2\pi)$, we have $u \in [0, 4\pi)$.

Solve $\cos u = -1$: $u = \pi, 3\pi$ (within $[0, 4\pi)$)

Convert back: $x = \frac{u}{2}$

$$x = \frac{\pi}{2}, \quad x = \frac{3\pi}{2}$$

$$x = \frac{\pi}{2}, \frac{3\pi}{2}$$

9. $\sqrt{3} \tan\left(\frac{x}{2}\right) - 1 = 0$ on $[0, 2\pi)$

Isolate: $\tan\left(\frac{x}{2}\right) = \frac{1}{\sqrt{3}}$

Let $u = \frac{x}{2}$. Since $x \in [0, 2\pi)$, we have $u \in [0, \pi)$.

Reference angle: $\frac{\pi}{6}$ (since $\tan \frac{\pi}{6} = \frac{1}{\sqrt{3}}$)

Tangent positive in QI only (within $[0, \pi)$): $u = \frac{\pi}{6}$

Convert back: $x = 2u = 2 \cdot \frac{\pi}{6} = \frac{\pi}{3}$

$$x = \frac{\pi}{3}$$

10. $\sin 2x = \frac{1}{2}$ on $[0, 2\pi)$

Let $u = 2x$. Since $x \in [0, 2\pi)$, we have $u \in [0, 4\pi)$.

Reference angle: $\frac{\pi}{6}$. Sine positive in QI and QII.

In $[0, 2\pi)$: $u = \frac{\pi}{6}, \frac{5\pi}{6}$

In $[2\pi, 4\pi)$: $u = 2\pi + \frac{\pi}{6} = \frac{13\pi}{6}, \quad u = 2\pi + \frac{5\pi}{6} = \frac{17\pi}{6}$

Convert back $x = \frac{u}{2}$:

$$x = \frac{\pi}{12}, \quad x = \frac{5\pi}{12}, \quad x = \frac{13\pi}{12}, \quad x = \frac{17\pi}{12}$$

$$x = \frac{\pi}{12}, \frac{5\pi}{12}, \frac{13\pi}{12}, \frac{17\pi}{12}$$